



GIRRAWEEEN HIGH SCHOOL

YEAR 12 HALF YEARLY EXAMINATION

1998

MATHEMATICS

3 UNIT

Time allowed - Two hours

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- All necessary working should be shown in every question.
- Marks may be deducted for careless or badly arranged work.
- Board-approved calculators may be used.
- Start each question on a new page.
- A table of standard integrals is provided.

Question 1. (18 marks)

(a) Find, correct to 1 decimal place:

(i) $\int_1^3 e^{2x} dx$

(ii) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2 x dx$

6

(b) Find the acute angle between the lines $y = 2x - 1$ and $x + y = 3$.
Give your answer correct to the nearest minute.

3

(c) Find an exact value for $\sin 75^\circ$.

3

(d) Sketch the curve $y = 2 \cos 3x$ for $0 \leq x \leq 2\pi$

3

(e) Solve $\sin 2x = \frac{\sqrt{3}}{2}$ for $0 \leq x \leq 2\pi$

3

Question 2 (24 marks)

(a) Use the substitution $u = x + 1$ to evaluate $\int_0^1 x(x+1)^4 dx$.

5

(b) It is known that 90% of students sitting for an examination will pass. In a random sample of 10 students sitting for this examination, what is the probability that exactly 3 students will fail. Give your answer correct to 2 decimal places.

3

(c) Show that $\frac{d}{dx}(x \log_e x) = \log_e x + 1$. Hence find $\int \log_e x dx$.

4

(d) Find the exact value of:

(i) $\cos(\cos^{-1}(\frac{1}{4}))$

(ii) $\sin^{-1}(\cos \frac{\pi}{3})$

(iii) $\tan\left[2 \tan^{-1}\left(\frac{1}{3}\right)\right]$

8

(e) (i) What is the domain and range of $y = \sin^{-1}\left(\frac{x}{3}\right)$

2

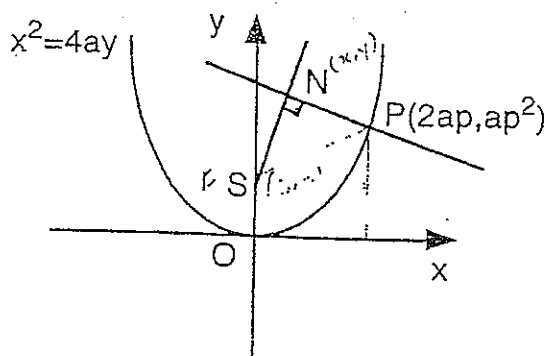
(ii) Sketch the curve $y = \sin^{-1}\left(\frac{x}{3}\right)$.

2

10-11-14

Question 3 (16 marks)

(a)



$P(2ap, ap^2)$ is a point on the parabola $x^2 = 4ay$.
 SN is perpendicular to the normal at P ,
 while $S(0, a)$ is the focus of the parabola

- (i) Show the equation of the normal is $x + py = 2ap + ap^3$. 3
- (ii) Find the equation of SN . 3
- (iii) Find the coordinates of N . 3
- (iv) Show the locus of N is the parabola $x^2 = a(y - a)$ 3

(b) How many different committees of 3 men and 4 women can be formed from 7 men and 8 women if:

- (i) there are no restrictions? 2
- (ii) a particular man has to be on the committee? 2

Question 4 (23 marks)

(a) Find $\frac{dy}{dx}$ if:

(i) $y = \frac{\sin x}{x}$

(ii) $y = \ln(\tan(x^2))$

(iii) $y = \sin^{-1}\left(\frac{x}{3}\right)$ 7

(b) Find:

(i) $\int \frac{2x}{2x^2 + 1} dx$

(ii) $\int \frac{\sin x}{\cos x} dx$

(iii) $\int xe^{x^2} dx$ 9

(c) Find the coefficient of x in the expansion $\left(x^2 - \frac{1}{x^3}\right)^{13}$ 3

(d) Prove $\frac{\sin 2\theta + 1}{\cos 2\theta} = \frac{\cos \theta + \sin \theta}{\cos \theta - \sin \theta}$ 4

1 - 2 cos 2θ
 2 sin 2θ - 1

Question 5 (14 marks)

- (a) (i) How many ways can the letters HELLO be arranged in a line?
 (ii) If one of these arrangements was picked at random, what is the probability that LL would be together?

2
2

- (b) Solve the equation $3x^3 - 17x^2 - 8x + 12 = 0$ given that the product of two of its roots is 4.

4

- (c) Find the area enclosed between $y = \sin x$ and the x axis between $x = \frac{\pi}{4}$ and $x = \pi$. Give an exact answer, showing units.

3

- (d) Find the exact value of $\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos^2 x dx$

3

Question 6 (15 marks)

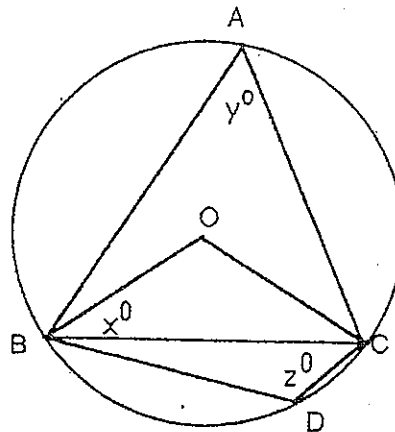
(a)

8

O is the centre of the circle

- (i) Prove that $x + y = 90$

- (ii) Prove that $z - y = 2x$



- (b) If $y = \tan^{-1} \frac{x}{1}$ prove that $2x \frac{dy}{dx} + (1 + x^2) \frac{d^2y}{dx^2} = 0$

3

- (c) The line $y = mx$ is tangent to the curve $y = e^{3x}$. Find m .

4

180-90-x

180-90-x

180-90-x

11

Question 1 19 marks

(a) (i) $\int_1^3 e^{2x} dx$

$$= \left[\frac{1}{2} e^{2x} \right]_1^3$$

$$= \frac{1}{2} [e^6 - e^2]$$

$\therefore = 198.0$

(3)

(ii) $\int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \sec^2 x dx = \left[\tan x \right]_{\frac{\pi}{6}}^{\frac{\pi}{4}}$

$$= 1 - \frac{1}{\sqrt{3}}$$

$= 0.4$

(3)

(b) $y = 2x - 1$ $m_1 = 2$

$y = -x + 3$ $m_2 = -1$

$\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$

$$= \left| \frac{2 - (-1)}{1 + (-2)} \right|$$

$$= \left| \frac{3}{-1} \right|$$

$\tan \theta = +3$

$\therefore \theta = 71^\circ 34'$

(3)

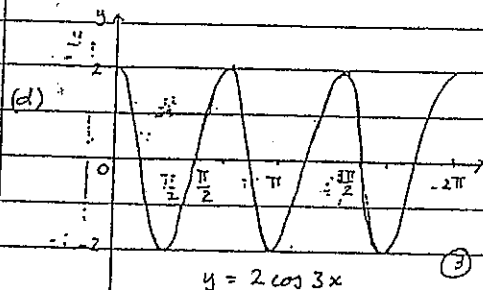
(c) $\sin(30^\circ + 45^\circ)$

$$= \sin 30^\circ \cos 45^\circ + \cos 30^\circ \sin 45^\circ$$

$$= \frac{1}{2} \times \frac{1}{\sqrt{2}} + \frac{\sqrt{3}}{2} \cdot \frac{1}{\sqrt{2}}$$

$$\sin 75^\circ = \frac{1 + \sqrt{3}}{2\sqrt{2}}$$

(3)



(e) $\sin 2x = \frac{\sqrt{3}}{2}$ $0 \leq x \leq 2\pi$

$0 \leq 2x \leq 4\pi$

$\therefore 2x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \frac{8\pi}{3}$

$\therefore x = \frac{\pi}{6}, \frac{\pi}{3}, \frac{7\pi}{6}, \frac{4\pi}{3}$

(3)

Question 2 24 marks

(a) $u = x + 1$ $x = 0, u = 1$

$\frac{du}{dx} = 1$ $x = 1, u = 2$

$\therefore du = dx$

$$\therefore \int_0^2 x(x+1)^4 dx = \int_1^2 (u-1) \cdot u^4 du$$

$$= \int_1^2 (u^5 - u^4) du$$

$$= \left[\frac{u^6}{6} - \frac{u^5}{5} \right]_1^2$$

$$= \left(\frac{64}{6} - \frac{32}{5} \right) - \left(\frac{1}{6} - \frac{1}{5} \right)$$

$$= \frac{64}{15} - \left(-\frac{1}{30} \right)$$

$$= \frac{43}{10} \text{ or } 4 \frac{3}{10}$$

(5)

(b) $p(\text{fail}) = 0.1$

$q(\text{not fail, but pass}) = 0.9$

$P(X=3) = {}^{10}C_3 (0.1)^3 (0.9)^7$

$= 0.06$

(3)

(c) $y = x \log_e x$

$$\frac{dy}{dx} = x \cdot \frac{1}{x} + \log_e x \cdot 1$$

$$= 1 + \log_e x$$

$$\therefore \int (1 + \log_e x) dx = x \log_e x + C$$

$$\int 1 dx + \int \log_e x dx = x \log_e x + C$$

$$x + \log_e x dx = x \log_e x + C$$

$$\therefore \int \log_e x dx = x \log_e x - x + C$$

(4)

(d) (i) $\cos(\cos^{-1}(\frac{1}{4})) = \frac{1}{4}$

(2)

(ii) $\sin^{-1}(\cos \frac{\pi}{3})$

$$= \sin^{-1}(\frac{1}{2})$$

$$= \frac{\pi}{6}$$

(2)

(iii) $\tan[2 \tan^{-1}(\frac{1}{3})]$

Let $\tan^{-1}(\frac{1}{3}) = d$

$$\therefore \tan 2d = \frac{2 \tan d}{1 - \tan^2 d}$$

$$= \frac{2 \cdot \tan(\tan^{-1}(\frac{1}{3}))}{1 - \tan^2(\tan^{-1}(\frac{1}{3}))}$$

$$= \frac{2 \cdot \frac{1}{3}}{1 - (\frac{1}{3})^2}$$

$$= \frac{2 \cdot \frac{1}{3}}{1 - \frac{1}{9}}$$

$$= \frac{\frac{2}{3}}{1 - \frac{1}{9}}$$

$$= \frac{3}{4}$$

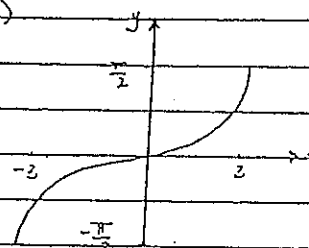
(4)

(e) (i)

Domain: $-3 \leq x \leq 3$

Range: $-\frac{\pi}{3} \leq y \leq \frac{\pi}{2}$

(ii)



(2)

Question 3 (16 marks)

(a) (i) $y = \frac{x^2}{4a}$

$$\frac{dy}{dx} = \frac{2x}{4a}$$

at $x = 2ap$ $\frac{dy}{dx} = p$

gradient of normal $= -\frac{1}{p}$

Equation of normal:

$$y - ap^2 = -\frac{1}{p}(x - 2ap)$$

$$py - ap^3 = -x + 2ap$$

$$\therefore x + py = 2ap + ap^3$$

(3)

(ii) SN: gradient is p

passes through (0, a)

Equation of SN:

$$y - a = p(x - 0)$$

$$px - y + a = 0$$

(3)

(iii) coordinates of N

$$y = px + a$$

$$\therefore x + p(px + a) = 2ap + ap^3$$

$$x + p^2x + ap = 2ap + ap^3$$

$$x(1 + p^2) = ap + ap^3$$

$$x(1+p^2) = op(1+p^2)$$

$$\therefore x = op$$

$$\therefore y = p(op) + a$$

$$y = ap^2 + a$$

$$\therefore N \text{ is } (ap, o(p^2+1)) \quad (1)$$

(iv) locus of N:

$$x = op \quad y = a(p^2+1)$$

$$\therefore p = \frac{x}{a}$$

$$\therefore y = a\left(\frac{x^2}{a^2} + 1\right)$$

$$\therefore y = \frac{x^2}{a} + a$$

$$ay = x^2 + a^2$$

$$\therefore x^2 = ay - a^2$$

$$x^2 = a(y-a) \quad (3)$$

(b) (i)

$$\text{Committees} = {}^7C_3 \times {}^2C_4$$

$$= 2450 \quad (2)$$

(ii) If one man is included:

$$\text{Committees} = {}^6C_2 \times {}^8C_4$$

$$= 1050 \quad (2)$$

Question 4 23 marks

(a)

$$(i) \frac{dy}{dx} = \frac{x \cos x - \sin x}{x^2} \quad (2)$$

$$(ii) \frac{dy}{dx} = \frac{1}{\tan(x)^4} \cdot \sec^2(x) \cdot 2x$$

$$= 2x \sec(x)^2$$

$$\tan(x)^2 \quad (3)$$

$$(iii) \frac{dy}{dx} = \frac{1}{\sqrt{9-x^2}} \quad (2)$$

$$(b) (i) \int \frac{2x}{2x^2+1} dx = \frac{1}{2} \int \frac{4x}{2x^2+1} dx$$

$$= \frac{1}{2} \ln(2x^2+1) + C \quad (3)$$

$$(ii) \int \frac{\sin x}{\cos x} dx = - \int \frac{-\sin x}{\cos x} dx$$

$$= -\ln \cos x + C \quad (3)$$

$$(iii) \int x e^x dx = \frac{1}{2} \int e^x \cdot 2x dx$$

$$= \frac{1}{2} e^x + C \quad (3)$$

(c)

$$\text{Term in } x^5$$

$$= {}^{13}C_5 (x^2)^8 \left(-\frac{1}{x^3}\right)^5$$

$$= {}^{13}C_5 \cdot x^{16} \cdot -\frac{1}{x^{15}}$$

$$= -{}^{13}C_5 x$$

$$\text{or } -1287x \quad (3)$$

$$(b) \frac{\sin 2\theta + 1}{\cos 2\theta}$$

$$= \frac{2 \sin \theta \cos \theta + \cos^2 \theta + \sin^2 \theta}{\cos^2 \theta + \sin^2 \theta}$$

$$= \frac{(\cos \theta + \sin \theta)^2}{(\cos \theta + \sin \theta)(\cos \theta + \sin \theta)}$$

$$= \frac{\cos \theta + \sin \theta}{\cos \theta + \sin \theta}$$

(4)

Question 5 (14 marks)

(a) (i)

$$\text{Arrangements} = \frac{5!}{2!}$$

$$= 60 \quad (2)$$

(ii) consider 11 as 1 letter.

$$\text{Arrangements} = 4! \quad (24)$$

$$\therefore \text{Probability} = \frac{24}{60} = \frac{2}{5} \quad (2)$$

$$(b) 3x^3 - 17x^2 - 8x + 12 = 0$$

$$\text{Roots } \alpha + \beta + \gamma = \frac{17}{3} \quad (4)$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = -\frac{8}{3} \quad (5)$$

$$\alpha\beta\gamma = -\frac{12}{3} = -4 \quad (6)$$

$$\text{Let } \alpha\beta = 4$$

$$\text{From (6)} \quad \alpha\beta\gamma = -4$$

$$4\gamma = -4$$

$$\gamma = -1$$

$\therefore$ one factor is $x+1$

$$3x^3 - 20x^2 + 12$$

$$x+1 \overline{) 3x^3 - 17x^2 - 8x + 12}$$

$$3x^3 + 3x^2$$

$$-20x^2 - 8x$$

$$-20x^2 - 20x$$

$$12x + 12$$

$$12x + 12$$

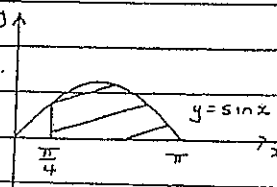
$$0$$

$$\therefore (x+1)(3x^2 - 20x + 12) = 0$$

$$(x+1)(3x-2)(x-6) = 0$$

Roots are: $-1, \frac{2}{3}$ and 6

(c)



$$A = \int_{\pi/4}^{\pi} \sin x dx$$

$$= [-\cos x]_{\pi/4}^{\pi}$$

$$= -[\cos \pi - \cos \frac{\pi}{4}]$$

$$= -[-1 - \frac{1}{\sqrt{2}}]$$

$$= 1 + \frac{1}{\sqrt{2}} \text{ unit}^2$$

$$(d) \cos 2x = 2\cos^2 x - 1$$

$$\frac{1}{2}(\cos 2x + 1) = \cos^2 x$$

$$\therefore \int_{\pi/4}^{\pi/2} \cos^2 x dx = \int_{\pi/4}^{\pi/2} \frac{1}{2}(\cos 2x + 1) dx$$

$$= \frac{1}{2} \left[\frac{\sin 2x}{2} + x \right]_{\pi/4}^{\pi/2}$$

$$= \frac{1}{2} \left[\left(\frac{\sin \pi}{2} + \frac{\pi}{2} \right) - \left(\frac{\sin \frac{\pi}{2}}{2} + \frac{\pi}{4} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{2} - \left(\frac{1}{2} + \frac{\pi}{4} \right) \right]$$

$$= \frac{1}{2} \left[\frac{\pi}{4} - \frac{1}{2} \right]$$

$$= \frac{1}{8} [\pi - 2]$$

$$\text{or } \frac{\pi}{8} - \frac{1}{4}$$

Q6

a) i) $OB = OC$ (radii of a circle)

$\therefore \triangle BOC$ is isosceles

$$\therefore \angle BCO = x^\circ$$

$\angle BOC = 2y$ ($\angle$ at the centre is twice of the circumference)

$$\therefore 2x + 2y = 180 \text{ (Angle sum of a } \triangle)$$

$$\therefore x + y = 90^\circ$$

ii) $2 + y = 180^\circ$ (opposite $\angle$ of a cyclic quadrilateral are supplementary)

$$\text{but } 2x + 2y = 180^\circ$$

$$2 + y = 2x + 2y$$

$$2 - y = 2x$$

$$b) y = \tan^{-1} x$$

$$y' = \frac{1}{1+x^2}$$

$$y'' = \frac{-2x}{(1+x^2)^2}$$

$$2x \left(\frac{1}{1+x^2} \right) + \cancel{(1+x^2)} \left(\frac{-2x}{(1+x^2)^2} \right) = 0$$

$$\frac{2x}{1+x^2} - \frac{2x}{1+x^2} = 0$$

$$0 = 0$$

$$\therefore LHS = RHS$$

$$c) y = mx$$

$$y = e^{3x}$$

$$m = 3e^{3x}$$

$$\therefore m = 3e^{3x}$$